GATE ECE

Previous Year Questions — Chapter-wise, with Answers and Solutions

Typeset edition

Papers included: GATE 2014 Set 1.

GATE 2014 Set 1

GATE 2014 (Set 1) | GA Q.1 | 1 mark

Choose the most appropriate phrase from the options given below to complete the following sentence.
The aircraft _____ take off as soon as its flight plan was filed.

(A) is allowed to
(B) will be allowed to
(C) was allowed to
(D) has been allowed to

Answer: (C) [answer matches official key]

Solution. Step 1. The clause "*was filed*" is in the simple past.
Step 2. The main clause must also be in the past: *was allowed to*.

GATE 2014 (Set 1) | GA Q.2 | 1 mark

Read the statements: All women are entrepreneurs. Some women are doctors. Which of the following conclusions can be logically inferred from the above statements?

(A) All women are doctors
(B) All doctors are entrepreneurs
(C) All entrepreneurs are women
(D) Some entrepreneurs are doctors

Answer: (D) [answer matches official key]

Solution. Step 1. Some women are doctors, and every woman is an entrepreneur.
Step 2. Hence those women who are doctors are also entrepreneurs: *some entrepreneurs are doctors*. The other statements are not forced.

GATE 2014 (Set 1) | GA Q.3 | 1 mark

Choose the most appropriate word from the options given below to complete the following sentence.
Many ancient cultures attributed disease to supernatural causes. However, modern science has largely helped _____ such notions.

(A) impel
(B) dispel
(C) propel
(D) repel

Answer: (B) [answer matches official key]

Solution. Step 1. The sentence says science has removed these ideas, i.e. *dispelled* them.

GATE 2014 (Set 1) | GA Q.4 | 1 mark

The statistics of runs scored in a series by four batsmen are provided in the following table. Who is the most consistent batsman of these four?

Batsman	Average	Standard deviation
K	31.2	5.21
L	46.0	6.35
M	54.4	6.22
N	17.9	5.90

(A) K
(B) L
(C) M
(D) N

Answer: (A) [answer matches official key]

Solution. Step 1. Consistency is measured by the spread of the scores: the batsman with the smallest standard deviation is the most consistent.
Step 2. The standard deviations are 5.21, 6.35, 6.22, 5.90; the smallest belongs to K.

GATE 2014 (Set 1) | GA Q.5 | 1 mark

What is the next number in the series?
12 35 81 173 357 _____

Answer: 725 [answer matches official key]

Solution. Step 1. Differences between consecutive terms: 23, 46, 92, 184 (each is double the previous).
Step 2. Equivalently $a_{n+1} = 2a_n + 11$.
Step 3. Next: $357 \times 2 + 11 = 725$.

GATE 2014 (Set 1) | GA Q.6 | 2 marks

Find the odd one from the following group:
W,E,K,O I,Q,W,A F,N,T,X N,V,B,D

(A) W,E,K,O
(B) I,Q,W,A
(C) F,N,T,X
(D) N,V,B,D

Answer: (D) [answer matches official key]

Solution. Step 1. In the first three groups the gaps between successive letters are $+8, +6, +4$ (e.g. W→E→K→O).
Step 2. In N,V,B,D the gaps are $+8, +6, +2$: the last gap breaks the pattern. So N,V,B,D is the odd one.

GATE 2014 (Set 1) | GA Q.7 | 2 marks

For submitting tax returns, all resident males with annual income below Rs 10 lakh should fill up Form P and all resident females with income below Rs 8 lakh should fill up Form Q. All people with incomes above Rs 10 lakh, except non residents with income above Rs 15 lakhs, should fill up Form R, while non residents with income above Rs 15 lakhs should fill up Form S. All others should fill Form T. An example of a person who should fill Form T is

(A) a resident male with annual income Rs 9 lakh
(B) a resident female with annual income Rs 9 lakh
(C) a non-resident male with annual income Rs 16 lakh
(D) a non-resident female with annual income Rs 16 lakh

Answer: (B) [answer matches official key]

Solution. Step 1. Check each person: (A) resident male < 10 lakh: Form P. (C), (D) non-residents > 15 lakh: Form S.
Step 2. (B) A resident female earning 9 lakh is not below 8 lakh (not Q) and not above 10 lakh (not R), so she falls under "all others": Form T.

GATE 2014 (Set 1) | GA Q.8 | 2 marks

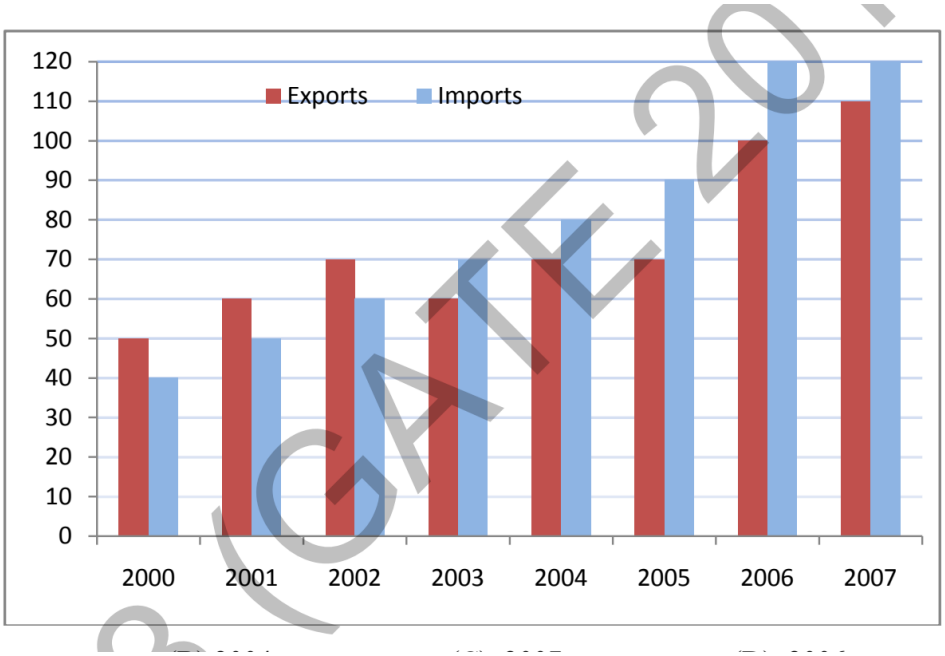
A train that is 280 metres long, travelling at a uniform speed, crosses a platform in 60 seconds and passes a man standing on the platform in 20 seconds. What is the length of the platform in metres?

Answer: 560 [answer matches official key]

Solution. Step 1. Passing the man means covering its own length: $v = \dfrac{280}{20} = 14$ m/s.
Step 2. Crossing the platform covers $280 + L$ in 60 s: $280 + L = 14 \times 60 = 840$.
Step 3. $L = 560$ m.

GATE 2014 (Set 1) | GA Q.9 | 2 marks

The exports and imports (in crores of Rs.) of a country from 2000 to 2007 are given in the following bar chart. If the trade deficit is defined as excess of imports over exports, in which year is the trade deficit 1/5th of the exports?



(A) 2005
(B) 2004
(C) 2007
(D) 2006

Answer: (D) [answer matches official key]

Solution. Step 1. Trade deficit = imports − exports; we need imports − exports $= \dfrac{1}{5}$exports, i.e. imports $= 1.2 \times$ exports.
Step 2. Reading the chart year by year, only 2006 satisfies imports $\approx 1.2\times$ exports.

GATE 2014 (Set 1) | GA Q.10 | 2 marks

You are given three coins: one has heads on both faces, the second has tails on both faces, and the third has a head on one face and a tail on the other. You choose a coin at random and toss it, and it comes up heads. The probability that the other face is tails is

(A) 1/4
(B) 1/3
(C) 1/2
(D) 2/3

Answer: (B) [answer matches official key]

Solution. Step 1. $P(\text{heads}) = \frac{1}{3}(1) + \frac{1}{3}(0) + \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{2}$.

Step 2. The other face is tails only for the mixed coin: $P(\text{mixed and heads}) = \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}$.

Step 3. $P = \frac{1/6}{1/2} = \frac{1}{3}$.

GATE 2014 (Set 1) | Q.1 | 1 mark

For matrices of same dimension M, N and scalar c, which one of these properties DOES NOT ALWAYS hold?

(A) $(M^T)^T = M$
(B) $(cM)^T = cM^T$
(C) $(M+N)^T = M^T + N^T$
(D) $MN = NM$

Answer: (D) [answer matches official key]

Solution. Step 1. (A), (B), (C) are standard properties of the transpose.
Step 2. Matrix multiplication is not commutative in general, so $MN = NM$ does not always hold.

GATE 2014 (Set 1) | Q.2 | 1 mark

In a housing society, half of the families have a single child per family, while the remaining half have two children per family. The probability that a child picked at random, has a sibling is _____.

Answer: 0.667 [answer matches official key]

Solution. Step 1. Take 2 families: one with 1 child and one with 2 children: 3 children in all.
Step 2. Only the 2 children of the second family have a sibling: $P = \frac{2}{3} = 0.667$.

GATE 2014 (Set 1) | Q.3 | 1 mark

C is a closed path in the z-plane given by $|z| = 3$. The value of the integral $\oint_C \left(\frac{z^2 - z + 4j}{z + 2j}\right) dz$ is

(A) $-4\pi(1 + j2)$
(B) $4\pi(3 - j2)$
(C) $-4\pi(3 + j2)$
(D) $4\pi(1 - j2)$

Answer: (C) [answer matches official key]

Solution. Step 1. The only singularity is the simple pole $z = -2j$, inside $|z| = 3$.
Step 2. Residue $= z^2 - z + 4j$ evaluated at $z = -2j$: $(-2j)^2 + 2j + 4j = -4 + 6j$.
Step 3. $\oint = 2\pi j(-4 + 6j) = -8\pi j - 12\pi = -4\pi(3 + j2)$.

GATE 2014 (Set 1) | Q.4 | 1 mark

A real (4×4) matrix A satisfies the equation $A^2 = I$, where I is the (4×4) identity matrix. The positive eigen value of A is _____.

Answer: 1 [answer matches official key]

Solution. Step 1. If $A\mathbf{v} = \lambda\mathbf{v}$ then $A^2\mathbf{v} = \lambda^2\mathbf{v} = \mathbf{v}$, so $\lambda^2 = 1$.
Step 2. $\lambda = \pm 1$; the positive eigenvalue is 1.

GATE 2014 (Set 1) | Q.5 | 1 mark

Let X_1, X_2, and X_3 be independent and identically distributed random variables with the uniform distribution on $[0, 1]$. The probability $P\{X_1 \text{ is the largest}\}$ is _____.

Answer: 0.333 [answer matches official key]

Solution. Step 1. The three variables are exchangeable and ties have probability 0.
Step 2. Each is equally likely to be the largest: $P = \dfrac{1}{3} = 0.333$.

GATE 2014 (Set 1) | Q.6 | 1 mark

For maximum power transfer between two cascaded sections of an electrical network, the relationship between the output impedance Z_1 of the first section to the input impedance Z_2 of the second section is

(A) $Z_2 = Z_1$
(B) $Z_2 = -Z_1$
(C) $Z_2 = Z_1^*$
(D) $Z_2 = -Z_1^*$

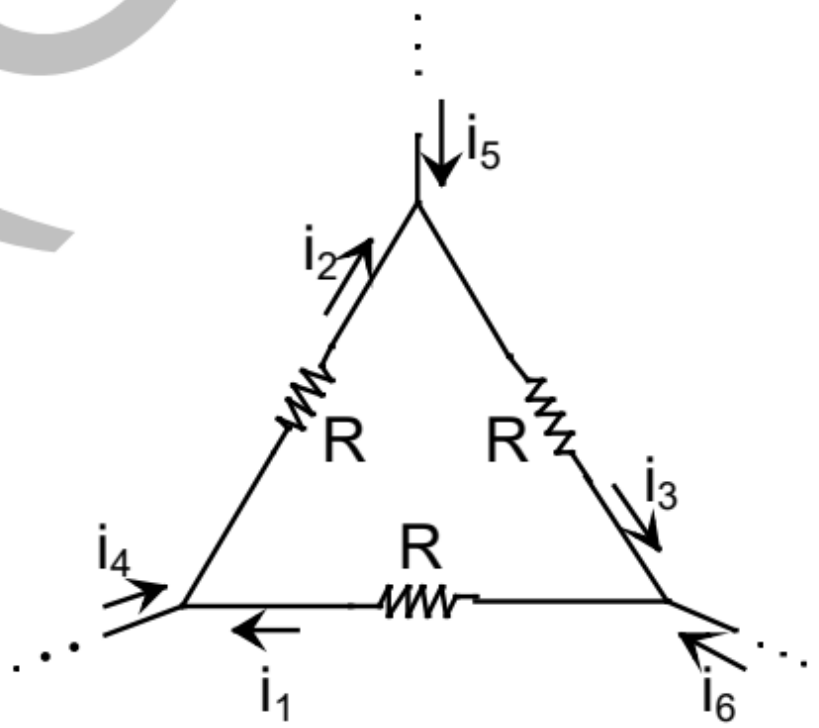
Answer: (C) [answer matches official key]

Solution. Step 1. Maximum power is delivered when the load impedance is the complex conjugate of the source impedance, which cancels the reactances and matches the resistances.
Step 2. $Z_2 = Z_1^*$.

GATE 2014 (Set 1) | Q.7 | 1 mark

Consider the configuration shown in the figure which is a portion of a larger electrical network. For $R = 1\ \Omega$ and currents $i_1 = 2$ A, $i_4 = -1$ A, $i_5 = -4$ A, which one of the following is TRUE?

in the figure which is a portion



(A) $i_6 = 5$ A
(B) $i_3 = -4$ A
(C) Data is sufficient to conclude that the supposed currents are impossible
(D) Data is insufficient to identify the currents i_2, i_3, i_6

Answer: (A) [answer matches official key]

Solution. **Step 1.** Draw a closed surface around the whole triangle; by KCL the net current entering the surface must be 0.
Step 2. The currents entering from the larger network through the external branches are $i_4 = -1$ A and $i_5 = -4$ A; the remaining one, i_6, must carry their sum away.
Step 3. $i_6 = -(i_4 + i_5) = -(-1 - 4) = 5$ A (the branch current i_1 is internal to the loop and does not affect this balance).

GATE 2014 (Set 1) | Q.8 | 1 mark

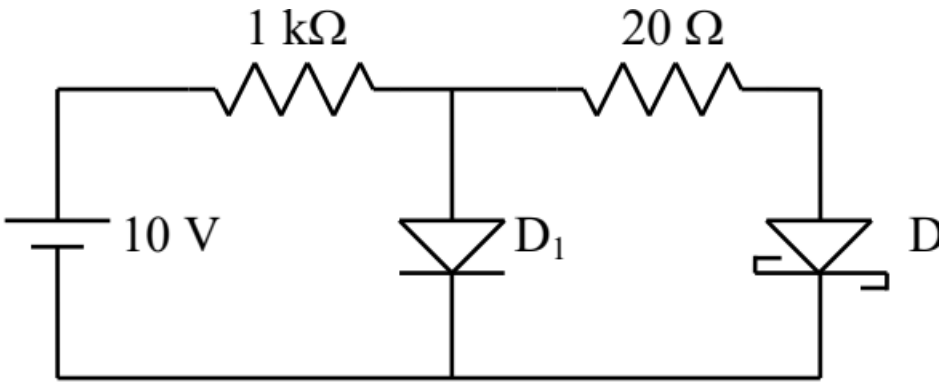
When the optical power incident on a photodiode is 10 μW and the responsivity is 0.8 A/W, the photocurrent generated (in μA) is _____.

Answer: 8 [answer matches official key]

Solution. **Step 1.** Photocurrent $= \mathcal{R}\, P_{opt}$.
Step 2. $I_{ph} = 0.8$ A/W $\times$ 10 μW $= 8$ μA.

GATE 2014 (Set 1) | Q.9 | 1 mark

In the figure, assume that the forward voltage drops of the PN diode D_1 and Schottky diode D_2 are 0.7 V and 0.3 V, respectively. If ON denotes conducting state of the diode and OFF denotes non-conducting state of the diode, then in the circuit,



(A) both D_1 and D_2 are ON
(B) D_1 is ON and D_2 is OFF
(C) both D_1 and D_2 are OFF
(D) D_1 is OFF and D_2 is ON

Answer: (D) [answer matches official key]

Solution. Step 1. The two diodes are connected in parallel across the same node voltage.
Step 2. The Schottky diode turns on at 0.3 V and then holds the common node at 0.3 V.
Step 3. At 0.3 V the PN diode (0.7 V needed) stays OFF: D_1 OFF, D_2 ON.

GATE 2014 (Set 1) | Q.10 | 1 mark

If fixed positive charges are present in the gate oxide of an n-channel enhancement type MOSFET, it will lead to

(A) a decrease in the threshold voltage
(B) channel length modulation
(C) an increase in substrate leakage current
(D) an increase in accumulation capacitance

Answer: (A) [answer matches official key]

Solution. Step 1. Positive oxide charge attracts electrons toward the oxide–semiconductor interface and so helps inversion.

Step 2. $V_T = V_{FB} + 2\phi_F + \dfrac{\sqrt{2\varepsilon q N_A (2\phi_F)}}{C_{ox}}$ with $V_{FB} = \phi_{ms} - \dfrac{Q_{ox}}{C_{ox}}$, so a positive Q_{ox} lowers V_T.

GATE 2014 (Set 1) | Q.11 | 1 mark

A good current buffer has

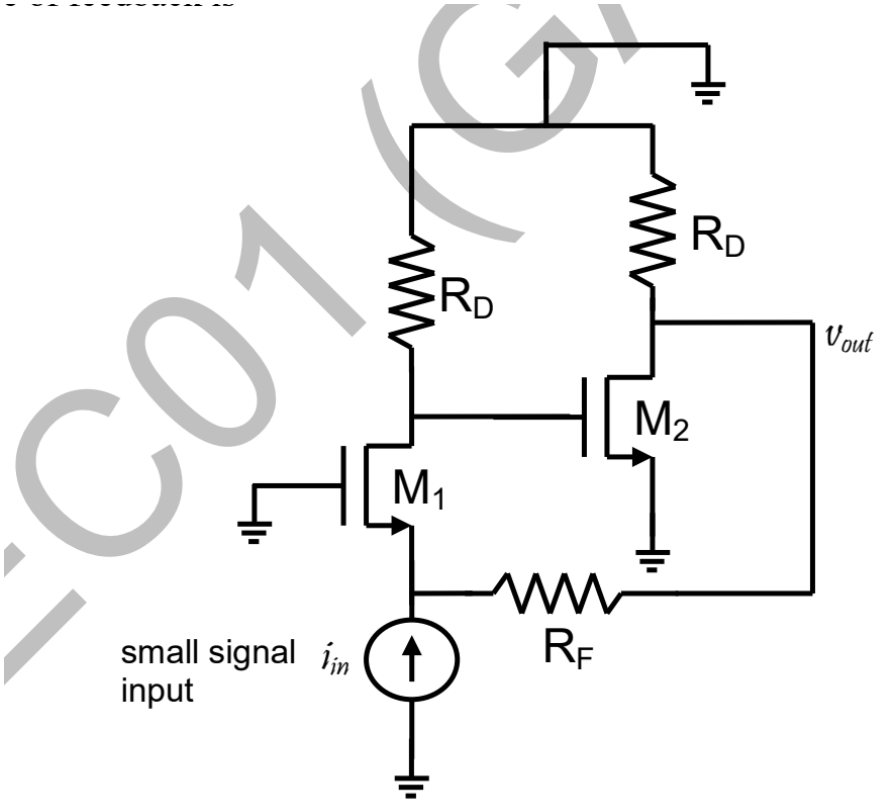
(A) low input impedance and low output impedance
(B) low input impedance and high output impedance
(C) high input impedance and low output impedance
(D) high input impedance and high output impedance

Answer: (B) [answer matches official key]

Solution. Step 1. A current buffer must accept the source current fully: input impedance much lower than the source impedance (ideally 0).
Step 2. It must deliver the current into any load: output impedance much higher than the load (ideally ∞).

GATE 2014 (Set 1) | Q.12 | 1 mark

In the ac equivalent circuit shown in the figure, if i_{in} is the input current and R_F is very large, the type of feedback is



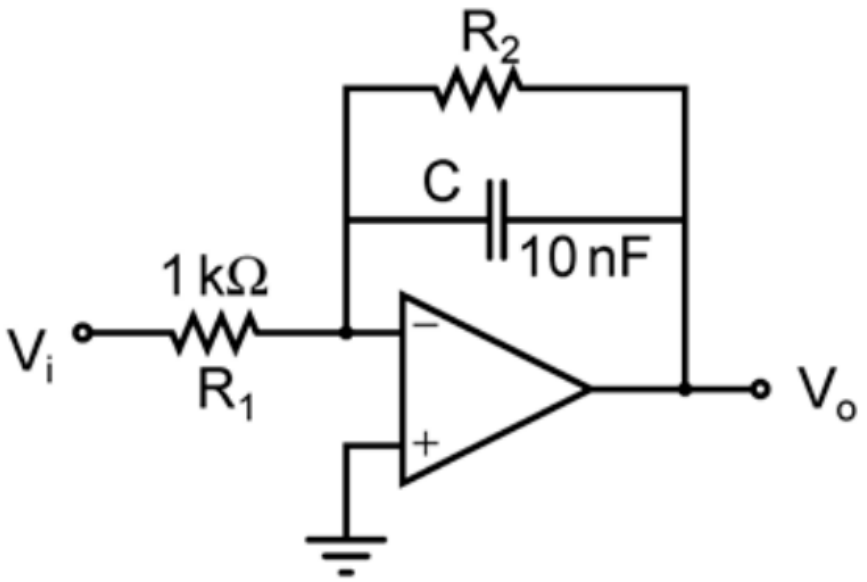
(A) voltage-voltage feedback
(B) voltage-current feedback
(C) current-voltage feedback
(D) current-current feedback

Answer: (B) [answer matches official key]

Solution. Step 1. The feedback resistor R_F connects the output node to the input (gate/source) node, and the input signal is a current source, so the feedback signal is mixed as a current (shunt connection at the input).
Step 2. The sampled quantity at the output is the output voltage v_{out} (taken in parallel with the output node): voltage sampling.
Step 3. Hence voltage-sampling, current-mixing feedback: "voltage–current" feedback.

GATE 2014 (Set 1) | Q.13 | 1 mark

In the low-pass filter shown in the figure, for a cut-off frequency of 5 kHz, the value of R_2 (in kΩ) is _____.



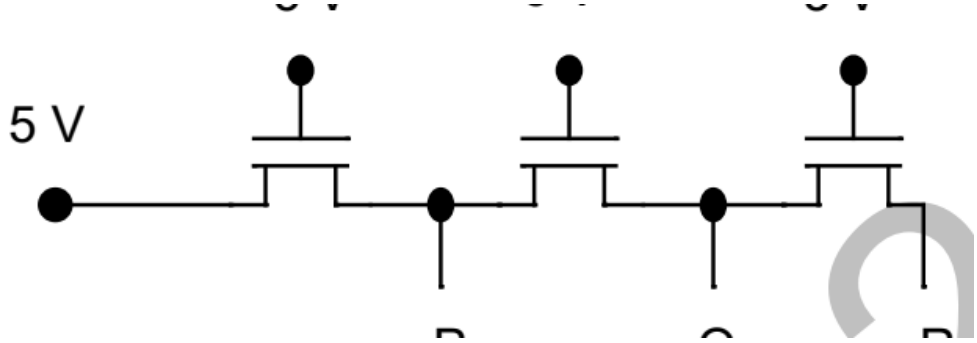
Answer: 3.18 [answer matches official key]

Solution. Step 1. The first-order active low-pass filter has $f_c = \dfrac{1}{2\pi R_2 C}$.

Step 2. $R_2 = \dfrac{1}{2\pi \times 5 \times 10^{3} \times 10 \times 10^{-9}} = 3.18$ kΩ.

GATE 2014 (Set 1) | Q.14 | 1 mark

In the following circuit employing pass transistor logic, all NMOS transistors are identical with a threshold voltage of 1 V. Ignoring the body-effect, the output voltages at P, Q and R are



(A) 4 V, 3 V, 2 V
(B) 5 V, 5 V, 5 V
(C) 4 V, 4 V, 4 V
(D) 5 V, 4 V, 3 V

Answer: (C) [answer matches official key]

Solution. Step 1. An NMOS pass transistor with gate at 5 V can pass a high level only up to $V_G - V_T = 5 - 1 = 4$ V.

Step 2. The first stage output $P = 4$ V. The next transistors also have gates at 5 V and source at 4 V, so they are already at their limit and do not reduce the voltage further.

Step 3. $P = Q = R = 4$ V.

GATE 2014 (Set 1) | Q.15 | 1 mark

The Boolean expression $(X+Y)(X+\overline{Y})+\overline{(X\overline{Y})+\overline{X}}$ simplifies to

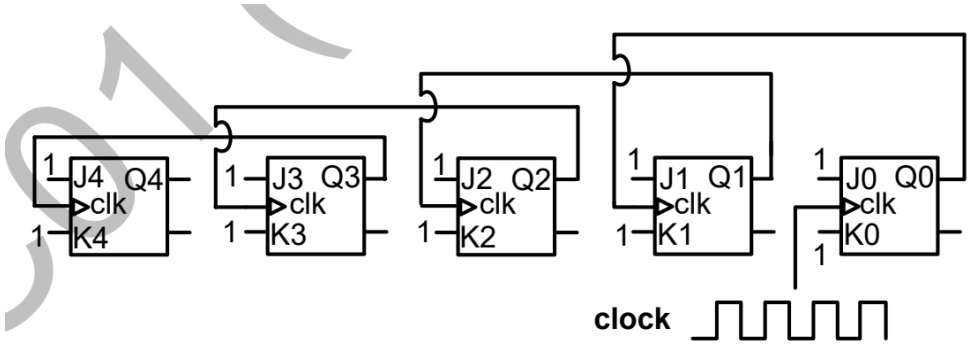
(A) X
(B) Y
(C) XY
(D) $X+Y$

Answer: (A) [answer matches official key]

Solution. Step 1. $(X+Y)(X+\overline{Y}) = X + Y\overline{Y} = X$.
Step 2. $\overline{X\overline{Y}+\overline{X}} = \overline{X\overline{Y}}\cdot X = (\overline{X}+Y)X = XY$.
Step 3. Sum: $X + XY = X$.

GATE 2014 (Set 1) | Q.16 | 1 mark

Five JK flip-flops are cascaded to form the circuit shown in Figure. Clock pulses at a frequency of 1 MHz are applied as shown. The frequency (in kHz) of the waveform at Q_3 is _____.



Answer: 62.5 [answer matches official key]

Solution. Step 1. With $J = K = 1$ every stage toggles; the feedback connections shown make the counter repeat after 16 clock pulses (the longest-cycle sequence reached by the feedback).
Step 2. The waveform at Q_3 is high for 8 and low for 8 clock periods, i.e. one cycle per 16 clocks.
Step 3. $f_{Q_3} = \dfrac{1\text{ MHz}}{16} = 62.5$ kHz.

GATE 2014 (Set 1) | Q.17 | 1 mark

A discrete-time signal $x[n] = \sin(\pi^2 n)$, n being an integer, is

(A) periodic with period π.
(B) periodic with period π^2.
(C) periodic with period $\pi/2$.
(D) not periodic.

Answer: (D) [answer matches official key]

Solution. Step 1. A discrete sinusoid $\sin(\Omega n)$ is periodic iff $\dfrac{\Omega}{2\pi}$ is rational.
Step 2. $\dfrac{\pi^2}{2\pi} = \dfrac{\pi}{2}$ is irrational, so $x[n]$ is not periodic.

GATE 2014 (Set 1) | Q.18 | 1 mark

Consider two real valued signals, $x(t)$ band-limited to $[-500$ Hz, 500 Hz$]$ and $y(t)$ band-limited to $[-1$ kHz, 1 kHz$]$. For $z(t) = x(t)\cdot y(t)$, the Nyquist sampling frequency (in kHz) is _____.

Answer: 3 [answer matches official key]

Solution. Step 1. Multiplication in time is convolution of spectra, so the bandwidths add: $500 + 1000 = 1500$ Hz.
Step 2. $f_s = 2 \times 1500 = 3000$ Hz $= 3$ kHz.

GATE 2014 (Set 1) | Q.19 | 1 mark

A continuous, linear time-invariant filter has an impulse response $h(t)$ described by $h(t) = \begin{cases} 3 & \text{for } 0 \le t \le 3 \\ 0 & \text{otherwise} \end{cases}$.
When a constant input of value 5 is applied to this filter, the steady-state output is _____.

Answer: 45 [answer matches official key]

Solution. Step 1. For a constant input $x = 5$ the steady-state output is $5 \int_{-\infty}^{\infty} h(t)\,dt$ (the dc gain times the input).
Step 2. $\int h\,dt = 3 \times 3 = 9$, so $y = 5 \times 9 = 45$.

GATE 2014 (Set 1) | Q.20 | 1 mark

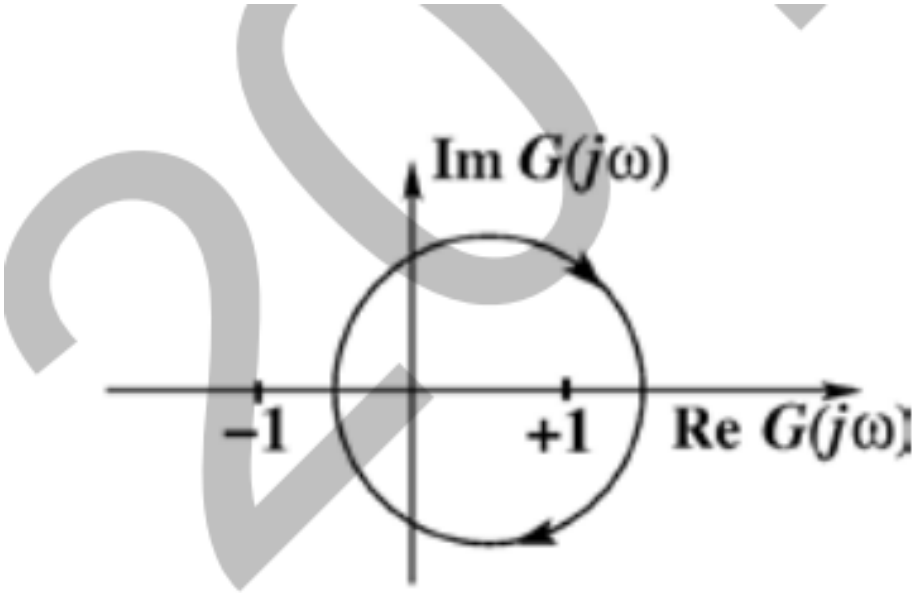
The forward path transfer function of a unity negative feedback system is given by $G(s) = \dfrac{K}{(s+2)(s-1)}$.
The value of K which will place both the poles of the closed-loop system at the same location, is _____.

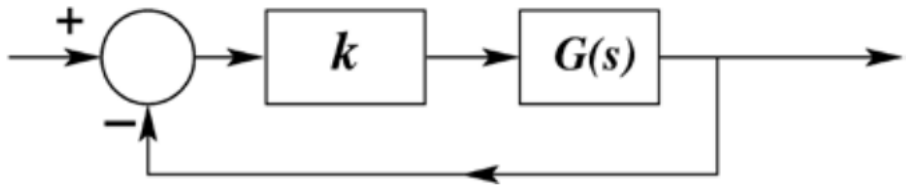
Answer: 2.25 [answer matches official key]

Solution. Step 1. Closed-loop characteristic equation: $(s+2)(s-1) + K = s^2 + s + (K-2) = 0$.
Step 2. Equal roots need a zero discriminant: $1 - 4(K-2) = 0$.
Step 3. $K = 2 + \dfrac{1}{4} = 2.25$ (both poles at $s = -0.5$).

GATE 2014 (Set 1) | Q.21 | 1 mark

Consider the feedback system shown in the figure. The Nyquist plot of $G(s)$ is also shown. Which one of the following conclusions is correct?





(A) $G(s)$ is an all-pass filter
(B) $G(s)$ is a strictly proper transfer function
(C) $G(s)$ is a stable and minimum-phase transfer function
(D) The closed-loop system is unstable for sufficiently large and positive k

Answer: (D) [answer matches official key]

Solution. Step 1. The loop gain is $kG(s)$, so the Nyquist plot of G is simply scaled by k.
Step 2. The plot encloses a region around the negative real axis; scaling by a large k makes the curve encircle the critical point -1 in the clockwise sense.
Step 3. By the Nyquist criterion the closed loop becomes unstable for sufficiently large positive k.

GATE 2014 (Set 1) | Q.22 | 1 mark

In a code-division multiple access (CDMA) system with $N = 8$ chips, the maximum number of users who can be assigned mutually orthogonal signature sequences is _____.

Answer: 8 [answer matches official key]

Solution. Step 1. Signature sequences are vectors of length $N = 8$ chips.
Step 2. At most N mutually orthogonal vectors exist in an N-dimensional space: 8 users.

GATE 2014 (Set 1) | Q.23 | 1 mark

The capacity of a Binary Symmetric Channel (BSC) with cross-over probability 0.5 is _____.

Answer: 0 [answer matches official key]

Solution. Step 1. $C = 1 - H_b(p)$ bits per use.
Step 2. For $p = 0.5$, $H_b(0.5) = 1$, so $C = 0$: the output is independent of the input.

GATE 2014 (Set 1) | Q.24 | 1 mark

A two-port network has scattering parameters given by $[S] = \begin{bmatrix} s_{11} & s_{12} \\ s_{21} & s_{22} \end{bmatrix}$. If the port-2 of the two-port is short circuited, the s_{11} parameter for the resultant one-port network is

(A) $\dfrac{s_{11} - s_{11}s_{22} + s_{12}s_{21}}{1 + s_{22}}$
(B) $\dfrac{s_{11} + s_{11}s_{22} - s_{12}s_{21}}{1 + s_{22}}$
(C) $\dfrac{s_{11} + s_{11}s_{22} + s_{12}s_{21}}{1 - s_{22}}$
(D) $\dfrac{s_{11} - s_{11}s_{22} + s_{12}s_{21}}{1 - s_{22}}$

Answer: (B) [answer matches official key]

Solution. Step 1. $b_1 = s_{11}a_1 + s_{12}a_2$ and $b_2 = s_{21}a_1 + s_{22}a_2$.

Step 2. A short circuit at port 2 reflects totally with $\Gamma = -1$: $a_2 = -b_2$, so $a_2 = -s_{21}a_1 - s_{22}a_2 \Rightarrow a_2 = -\dfrac{s_{21}}{1+s_{22}}a_1$.

Step 3. $s'_{11} = \dfrac{b_1}{a_1} = s_{11} - \dfrac{s_{12}s_{21}}{1+s_{22}} = \dfrac{s_{11} + s_{11}s_{22} - s_{12}s_{21}}{1+s_{22}}$.

GATE 2014 (Set 1) | Q.25 | 1 mark

The force on a point charge $+q$ kept at a distance d from the surface of an infinite grounded metal plate in a medium of permittivity ϵ is

(A) 0

(B) $\dfrac{q^2}{16\pi\epsilon d^2}$ away from the plate

(C) $\dfrac{q^2}{16\pi\epsilon d^2}$ towards the plate

(D) $\dfrac{q^2}{4\pi\epsilon d^2}$ towards the plate

Answer: (C) [answer matches official key]

Solution. Step 1. By the method of images the grounded plate is replaced by an image charge $-q$ at distance d behind the plate: separation $2d$.

Step 2. $F = \dfrac{q^2}{4\pi\epsilon(2d)^2} = \dfrac{q^2}{16\pi\epsilon d^2}$, attractive (towards the plate).

GATE 2014 (Set 1) | Q.26 | 2 marks

The Taylor series expansion of $3\sin x + 2\cos x$ is

(A) $2 + 3x - x^2 - \dfrac{x^3}{2} + \cdots$

(B) $2 - 3x + x^2 - \dfrac{x^3}{2} + \cdots$

(C) $2 + 3x + x^2 + \dfrac{x^3}{2} + \cdots$

(D) $2 - 3x - x^2 + \dfrac{x^3}{2} + \cdots$

Answer: (A) [answer matches official key]

Solution. Step 1. $3\sin x = 3x - \dfrac{x^3}{2} + \cdots$.

Step 2. $2\cos x = 2 - x^2 + \cdots$.

Step 3. Sum: $2 + 3x - x^2 - \dfrac{x^3}{2} + \cdots$.

GATE 2014 (Set 1) | Q.27 | 2 marks

For a function $g(t)$, it is given that $\displaystyle\int_{-\infty}^{+\infty} g(t)e^{-j\omega t}dt = \omega e^{-2\omega^2}$ for any real value ω. If $y(t) = \int_{-\infty}^{t} g(\tau)\,d\tau$, then $\displaystyle\int_{-\infty}^{+\infty} y(t)\,dt$ is

(A) 0

(B) $-j$

(C) $-\dfrac{j}{2}$

(D) $\dfrac{j}{2}$

Answer: (B) [answer matches official key]

Solution. Step 1. $G(\omega) = \omega e^{-2\omega^2}$, $G(0) = 0$. Integration in time divides the transform by $j\omega$: $Y(\omega) = \dfrac{G(\omega)}{j\omega}$ (no impulse term since $G(0) = 0$).

Step 2. $\int y\,dt = Y(0) = \lim_{\omega\to 0} \dfrac{\omega e^{-2\omega^2}}{j\omega} = \dfrac{1}{j} = -j.$

GATE 2014 (Set 1) | Q.28 | 2 marks

The volume under the surface $z(x, y) = x + y$ and above the triangle in the x-y plane defined by $\{0 \le y \le x$ and $0 \le x \le 12\}$ is _____.

Answer: 864 [answer matches official key]

Solution. Step 1. $V = \int_0^{12}\int_0^{x} (x + y)\,dy\,dx.$

Step 2. Inner integral: $x \cdot x + \dfrac{x^2}{2} = \dfrac{3}{2}x^2.$

Step 3. $V = \int_0^{12} \dfrac{3}{2}x^2 dx = \dfrac{1}{2}(12)^3 = 864.$

GATE 2014 (Set 1) | Q.29 | 2 marks

Consider the matrix $J_6 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ which is obtained by reversing the order of the columns of the identity matrix I_6. Let $P = I_6 + \alpha J_6$, where α is a non-negative real number. The value of α for which $\det(P) = 0$ is _____.

Answer: 1 [answer matches official key]

Solution. Step 1. $J_6^2 = I$, so the eigenvalues of J_6 are ± 1 (three of each).

Step 2. The eigenvalues of P are $1 \pm \alpha$.

Step 3. $\det P = 0$ requires $1 - \alpha = 0$: $\alpha = 1$.

GATE 2014 (Set 1) | Q.30 | 2 marks

A Y-network has resistances of 10 Ω each in two of its arms, while the third arm has a resistance of 11 Ω. In the equivalent Δ-network, the lowest value (in Ω) among the three resistances is _____.

Answer: 29.09 [answer matches official key]

Solution. Step 1. $Y \to \Delta$: $R_{ab} = \dfrac{R_1R_2 + R_2R_3 + R_3R_1}{R_{opposite}}$. The numerator is $10 \cdot 10 + 10 \cdot 11 + 11 \cdot 10 = 320$.

Step 2. The Δ resistances are $\dfrac{320}{10} = 32$, 32 and $\dfrac{320}{11} = 29.09$.

Step 3. The lowest is 29.09 Ω.

GATE 2014 (Set 1) | Q.31 | 2 marks

A 230 V rms source supplies power to two loads connected in parallel. The first load draws 10 kW at 0.8 leading power factor and the second one draws 10 kVA at 0.8 lagging power factor. The complex power delivered by the source is

(A) $(18 + j1.5)$ kVA
(B) $(18 - j1.5)$ kVA
(C) $(20 + j1.5)$ kVA

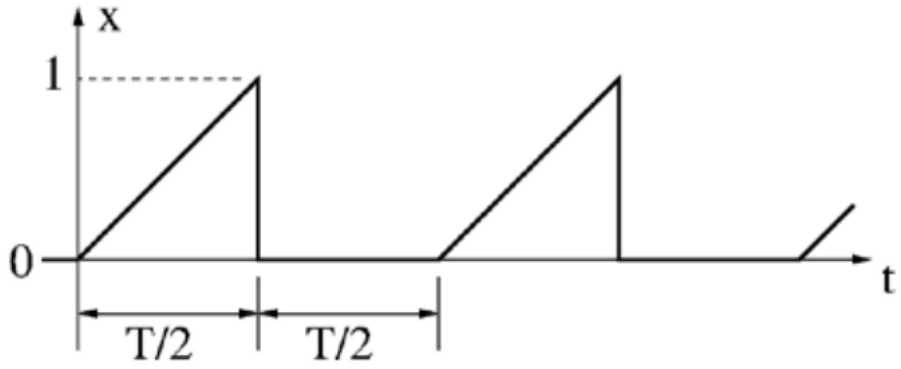
(D) $(20 - j1.5)$ kVA

Answer: (B) [answer matches official key]

Solution. Step 1. Load 1: $P_1 = 10$ kW at pf $= 0.8$ leading: $S_1 = 12.5$ kVA, $Q_1 = -7.5$ kVAR (capacitive).
Step 2. Load 2: $S_2 = 10$ kVA at 0.8 lagging: $P_2 = 8$ kW, $Q_2 = +6$ kVAR.
Step 3. $S = (10 + 8) + j(-7.5 + 6) = 18 - j1.5$ kVA.

GATE 2014 (Set 1) | Q.32 | 2 marks

A periodic variable x is shown in the figure as a function of time. The root-mean-square (rms) value of x is _____.

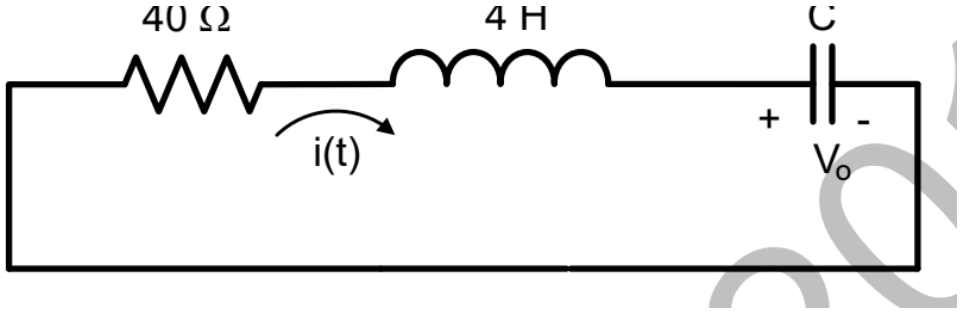


Answer: 0.408 [answer matches official key]

Solution. Step 1. In each period x rises linearly from 0 to 1 during $T/2$ and is 0 during the other $T/2$: $x = \dfrac{2t}{T}$ for $0 < t < T/2$.
Step 2. $\overline{x^2} = \dfrac{1}{T}\int_0^{T/2} \dfrac{4t^2}{T^2} dt = \dfrac{4}{T^3} \cdot \dfrac{T^3}{24} = \dfrac{1}{6}$.
Step 3. $x_{rms} = \dfrac{1}{\sqrt{6}} = 0.408$.

GATE 2014 (Set 1) | Q.33 | 2 marks

In the circuit shown in the figure, the value of capacitor C (in mF) needed to have critically damped response $i(t)$ is _____.



Answer: 10 [answer matches official key]

Solution. Step 1. Series RLC: critical damping when $R = 2\sqrt{L/C}$.
Step 2. $40 = 2\sqrt{4/C} \Rightarrow \sqrt{4/C} = 20 \Rightarrow C = \dfrac{4}{400} = 0.01$ F $= 10$ mF.

GATE 2014 (Set 1) | Q.34 | 2 marks

A BJT is biased in forward active mode. Assume $V_{BE} = 0.7$ V, $kT/q = 25$ mV and reverse saturation current $I_S = 10^{-13}$ A. The transconductance of the BJT is _____ (in mA/V).

Answer: 5.78 [answer matches official key]

Solution. Step 1. $I_C = I_S e^{V_{BE}/V_T} = 10^{-13} e^{28} = 0.1446$ A.

Step 2. $g_m = \dfrac{I_C}{V_T} = \dfrac{0.1446}{0.025} = 5.78$ A/V $(= 5780$ mA/V).

GATE 2014 (Set 1) | Q.35 | 2 marks

The doping concentrations on the p-side and n-side of a silicon diode are 1×10^{16} cm^{-3} and 1×10^{17} cm^{-3}, respectively. A forward bias of 0.3 V is applied to the diode. At $T = 300$ K, the intrinsic carrier concentration of silicon $n_i = 1.5 \times 10^{10}$ cm^{-3} and $\dfrac{kT}{q} = 26$ mV. The electron concentration at the edge of the depletion region on the p-side is

(A) 2.3×10^{9} cm^{-3}
(B) 1×10^{16} cm^{-3}
(C) 1×10^{17} cm^{-3}
(D) 2.25×10^{6} cm^{-3}

Answer: (A) [answer matches official key]

Solution. Step 1. Equilibrium minority electrons on the p-side: $n_{p0} = \dfrac{n_i^2}{N_A} = \dfrac{2.25 \times 10^{20}}{10^{16}} = 2.25 \times 10^{4}$ cm^{-3}.

Step 2. Under forward bias: $n_p = n_{p0} e^{V/V_T} = 2.25 \times 10^{4}\, e^{0.3/0.026} = 2.25 \times 10^{4} \times 1.03 \times 10^{5} = 2.3 \times 10^{9}$ cm^{-3}.

GATE 2014 (Set 1) | Q.36 | 2 marks

A depletion type N-channel MOSFET is biased in its linear region for use as a voltage controlled resistor. Assume threshold voltage $V_{TH} = -0.5$ V, $V_{GS} = 2.0$ V, $V_{DS} = 5$ V, $W/L = 100$, $C_{ox} = 10^{-8}$ F/cm^2 and $\mu_n = 800$ cm^2/V-s. The value of the resistance of the voltage controlled resistor (in Ω) is _____.

Answer: 500 [answer matches official key]

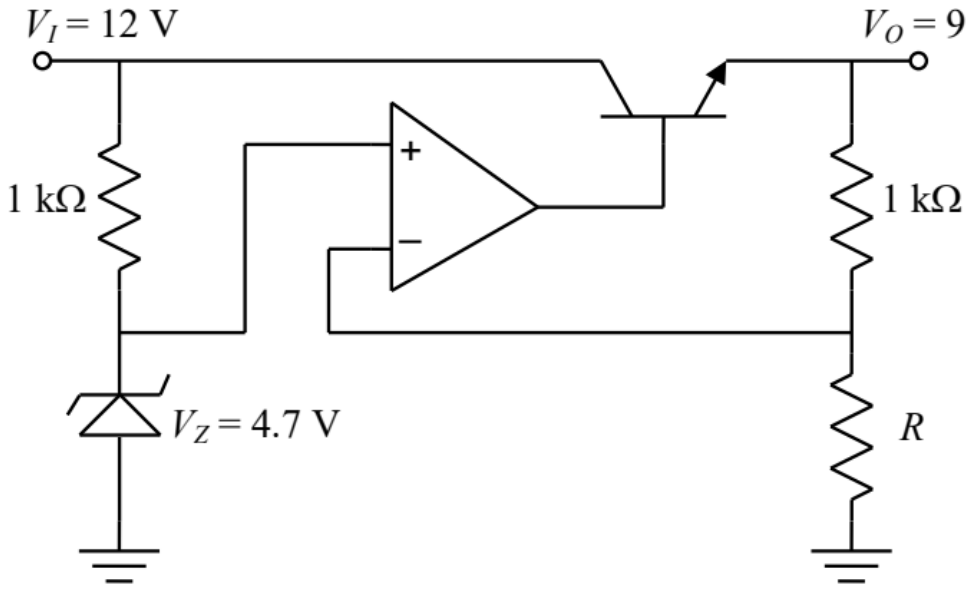
Solution. Step 1. For small V_{DS} in the linear region $I_D \approx \mu_n C_{ox} \dfrac{W}{L}(V_{GS} - V_{TH})V_{DS}$, so $r = \dfrac{1}{\mu_n C_{ox} \frac{W}{L}(V_{GS} - V_{TH})}$.

Step 2. $\mu_n C_{ox} \dfrac{W}{L} = 800 \times 10^{-8} \times 100 = 8 \times 10^{-4}$ A/V^2; $V_{GS} - V_{TH} = 2.5$ V.

Step 3. $r = \dfrac{1}{8 \times 10^{-4} \times 2.5} = 500\ \Omega$.

GATE 2014 (Set 1) | Q.37 | 2 marks

In the voltage regulator circuit shown in the figure, the op-amp is ideal. The BJT has $V_{BE} = 0.7$ V and $\beta = 100$, and the zener voltage is 4.7 V. For a regulated output of 9 V, the value of R (in Ω) is _____.



Answer: 1093 [answer matches official key]

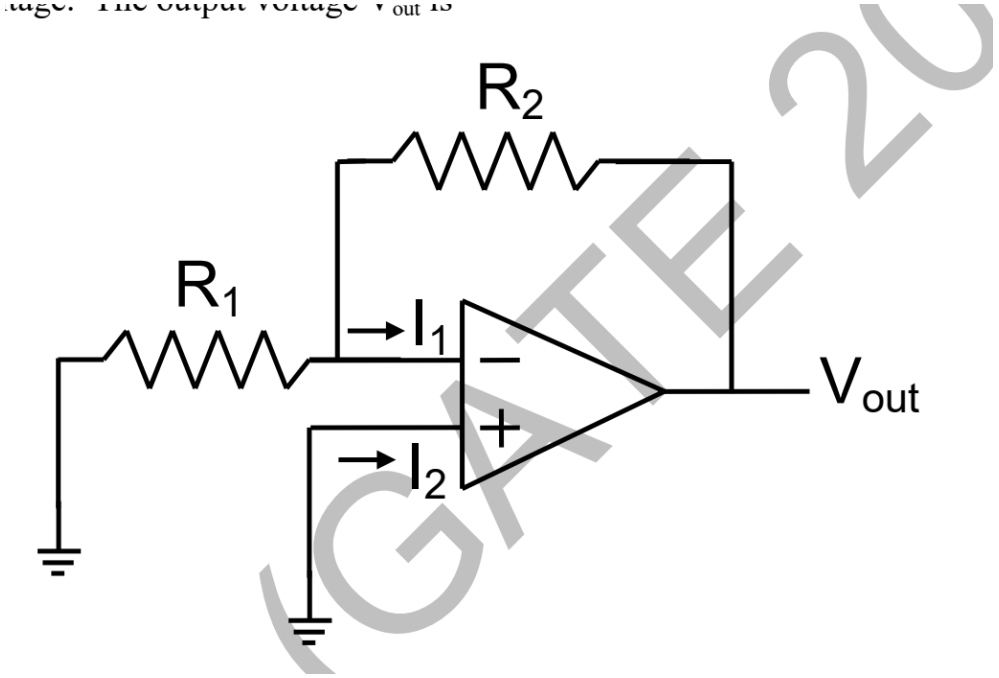
Solution. Step 1. The zener branch fixes the reference at $V_Z = 4.7$ V at the op-amp input.

Step 2. The feedback divider is R (lower) and 1 $extk\Omega$ (upper) across the 9 V output: the tap voltage is $9 \cdot \dfrac{R}{1000 + R}$.

Step 3. The ideal op-amp makes the tap equal to V_Z: $9R = 4.7(1000 + R) \Rightarrow 4.3R = 4700 \Rightarrow R = 1093\ \Omega$.

GATE 2014 (Set 1) | Q.38 | 2 marks

In the circuit shown, the op-amp has finite input impedance, infinite voltage gain and zero input offset voltage. The output voltage V_{out} is



(A) $-I_2(R_1 + R_2)$
(B) $I_2 R_2$
(C) $I_1 R_2$
(D) $-I_1(R_1 + R_2)$

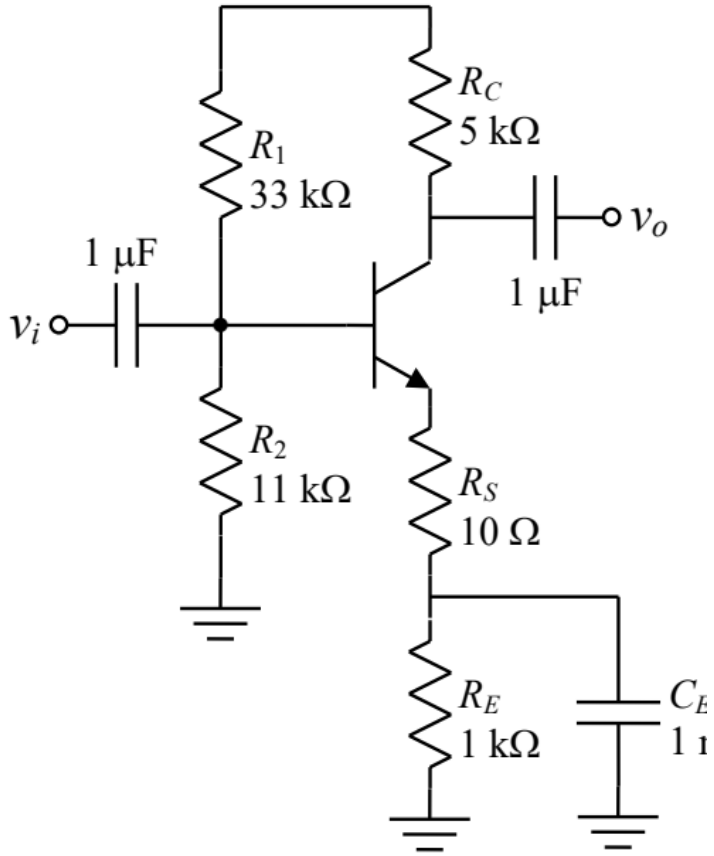
GATE 2

Answer: (C) [answer matches official key]

Solution. Step 1. Infinite gain forces the differential input to 0: the inverting terminal sits at the same voltage as the non-inverting terminal.
Step 2. The op-amp draws input currents I_1 (inverting) and I_2 (non-inverting); the non-inverting terminal is grounded through its source so $V_+ = 0$, hence $V_- = 0$.
Step 3. The current I_1 flows through R_2 from the output to the virtual ground: $V_{out} = I_1 R_2$.

GATE 2014 (Set 1) | Q.39 | 2 marks

For the amplifier shown in the figure, the BJT parameters are $V_{BE} = 0.7$ V, $\beta = 200$, and thermal voltage $V_T = 25$ mV. The voltage gain (v_o/v_i) of the amplifier is _____.

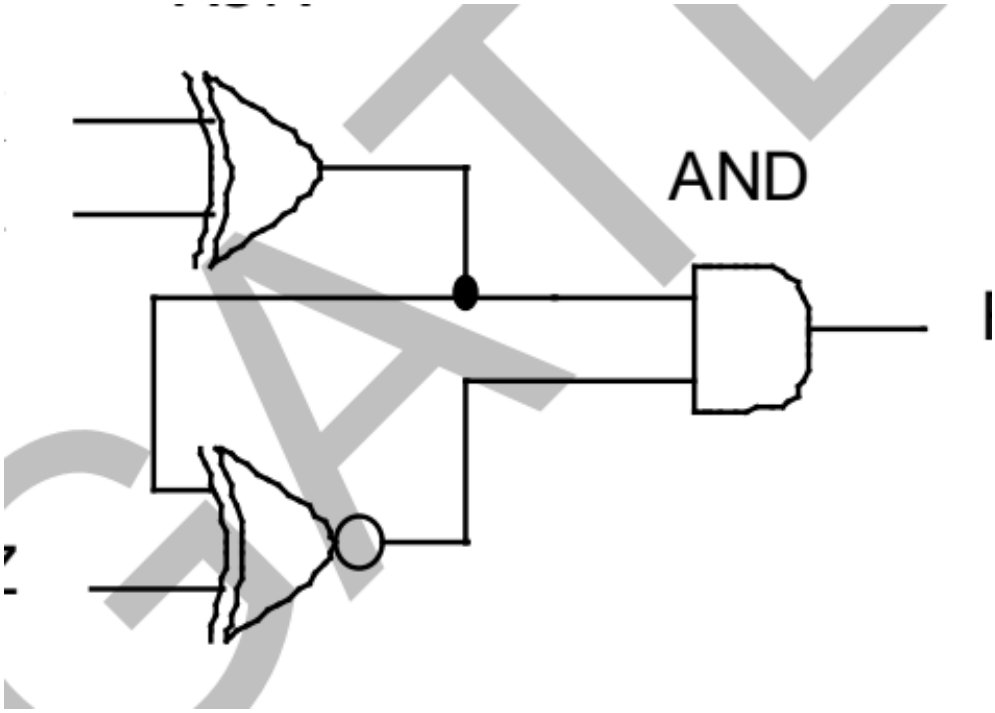


Answer: -235 [answer matches official key]

Solution. Step 1. $V_B = 12 \cdot \dfrac{11}{44} = 3$ V (Thevenin $R_B = 33 \| 11 = 8.25$ kΩ). $V_E = 2.3$ V; with $R_E = 1$ kΩ $+ 10$ Ω the emitter current is $I_E \approx \dfrac{3 - 0.7}{1.01\text{k} + R_B/201} = 2.2$ mA.
Step 2. $g_m = \dfrac{I_C}{V_T} \approx \dfrac{2.2}{25} = 88$ mS, $r_e \approx 11.4$ Ω.
Step 3. C_E bypasses the 1 kΩ, leaving only the 10 Ω resistor in the emitter: $A_v = -\dfrac{R_C}{r_e + 10} = -\dfrac{5000}{21.4} = -234 \approx -235$.

GATE 2014 (Set 1) | Q.40 | 2 marks

The output F in the digital logic circuit shown in the figure is



(A) $F = \overline{X}YZ + X\overline{Y}Z$
(B) $F = \overline{X}Y\,\overline{Z} + X\overline{Y}\,\overline{Z}$
(C) $F = \overline{X}\,\overline{Y}Z + XYZ$
(D) $F = \overline{X}\,\overline{Y}\,\overline{Z} + XY\overline{Z}$

Answer: (A) [answer matches official key]

Solution. Step 1. The XOR gate gives $X \oplus Y$ and the XNOR gate (inputs Z and the XOR output path) gives the complement with respect to Z; the AND combines them.
Step 2. $F = (X \oplus Y) \cdot Z$, i.e. $F = \overline{X}YZ + X\overline{Y}Z$.

GATE 2014 (Set 1) | Q.41 | 2 marks

Consider the Boolean function, $F(w, x, y, z) = wy + xy + \overline{w}\,xyz + \overline{w}\,x\,y + xz + \overline{x}\,\overline{y}\,z$. Which one of the following is the complete set of essential prime implicants?

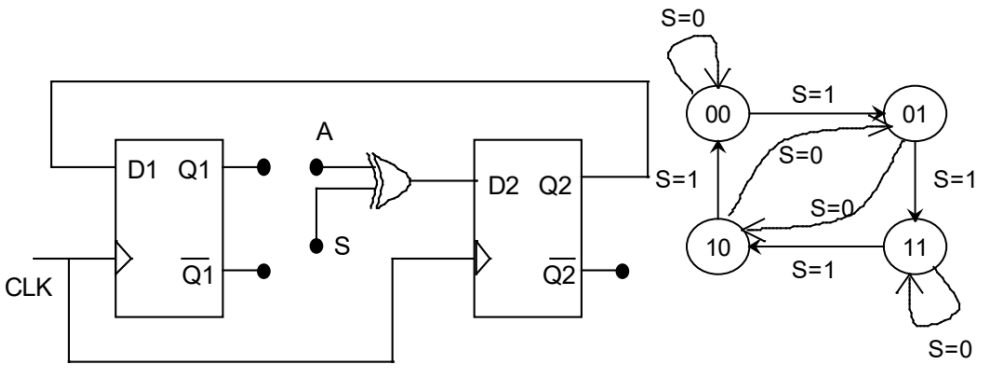
(A) $w,\ y,\ xz,\ \overline{x}\,\overline{z}$
(B) $w,\ y,\ xz$
(C) y
(D) $y,\ xz,\ \overline{x}\,\overline{z}$

Answer: (D) [answer matches official key]

Solution. Step 1. Simplify: terms with y ($wy + xy + \overline{w}xy + \ldots$) merge into the prime implicant y plus xz; the remaining minterms with $y = 0$ require $\overline{x}\,\overline{z}$.
Step 2. Verifying on the K-map, the essential prime implicants are y, xz and $\overline{x}\,\overline{z}$.

GATE 2014 (Set 1) | Q.42 | 2 marks

The digital logic shown in the figure satisfies the given state diagram when Q_1 is connected to input A of the XOR gate. Suppose the XOR gate is replaced by an XNOR gate. Which one of the following options preserves the state diagram?



(A) Input A is connected to $\overline{Q_2}$
(B) Input A is connected to Q_2
(C) Input A is connected to $\overline{Q_1}$ and S is complemented
(D) Input A is connected to $\overline{Q_1}$

Answer: (D) [answer matches official key]

Solution. Step 1. $\text{XNOR}(A, S) = \text{XOR}(\overline{A}, S)$.
Step 2. To keep the same logic function after replacing XOR by XNOR, the input A must be complemented: connect A to $\overline{Q_1}$.

GATE 2014 (Set 1) | Q.43 | 2 marks

Let $x[n] = \left(-\frac{1}{9}\right)^n u(n) - \left(-\frac{1}{3}\right)^n u(-n-1)$. The Region of Convergence (ROC) of the z-transform of $x[n]$

(A) is $|z| > \frac{1}{9}$.
(B) is $|z| < \frac{1}{3}$.
(C) is $\frac{1}{3} > |z| > \frac{1}{9}$.
(D) does not exist.

Answer: (C) [answer matches official key]

Solution. Step 1. The right-sided term has ROC $|z| > \frac{1}{9}$; the left-sided term has ROC $|z| < \frac{1}{3}$.
Step 2. Intersection: $\frac{1}{9} < |z| < \frac{1}{3}$.

GATE 2014 (Set 1) | Q.44 | 2 marks

Consider a discrete time periodic signal $x[n] = \sin\left(\frac{\pi n}{5}\right)$. Let a_k be the complex Fourier series coefficients of $x[n]$. The coefficients $\{a_k\}$ are non-zero when $k = Bm \pm 1$, where m is any integer. The value of B is _____.

Answer: 10 [answer matches official key]

Solution. Step 1. $x[n] = \sin\left(\frac{2\pi n}{10}\right)$ has period $N = 10$.
Step 2. The coefficients are non-zero at $k = \pm 1 \pmod{10}$: $k = 10m \pm 1$.
Step 3. $B = 10$.

GATE 2014 (Set 1) | Q.45 | 2 marks

A system is described by the following differential equation, where $u(t)$ is the input to the system and $y(t)$ is the output of the system. $\dot{y}(t) + 5y(t) = u(t)$. When $y(0) = 1$ and $u(t)$ is a unit step function, $y(t)$ is

(A) $0.2 + 0.8e^{-5t}$
(B) $0.2 - 0.2e^{-5t}$
(C) $0.8 + 0.2e^{-5t}$
(D) $0.8 - 0.8e^{-5t}$

Answer: (A) [answer matches official key]

Solution. Step 1. Steady state: $5y = 1 \Rightarrow y_{ss} = 0.2$. Natural response: Ce^{-5t}.
Step 2. $y(0) = 0.2 + C = 1 \Rightarrow C = 0.8$.
Step 3. $y(t) = 0.2 + 0.8e^{-5t}$.

GATE 2014 (Set 1) | Q.46 | 2 marks

Consider the state space model of a system, as given below: $\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix} = \begin{bmatrix}-1 & 1 & 0\\ 0 & -1 & 0\\ 0 & 0 & -2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}0\\ 4\\ 0\end{bmatrix} u;\ y = [1\ \ 1\ \ 1]\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$. The system is

(A) controllable and observable
(B) uncontrollable and observable
(C) uncontrollable and unobservable
(D) controllable and unobservable

Answer: (B) [answer matches official key]

Solution. Step 1. $B = (0, 4, 0)^T$, $AB = (4, -4, 0)^T$, $A^2B = (-8, 4, 0)^T$. The third row of $[B\ AB\ A^2B]$ is zero, so x_3 is uncontrollable.
Step 2. $C = [1\ 1\ 1]$, $CA = [-1\ 0\ -2]$, $CA^2 = [1\ -1\ 4]$; the matrix $\begin{bmatrix}1 & 1 & 1\\ -1 & 0 & -2\\ 1 & -1 & 4\end{bmatrix}$ has determinant $-3 \neq 0$: observable.
Step 3. Hence uncontrollable but observable.

GATE 2014 (Set 1) | Q.47 | 2 marks

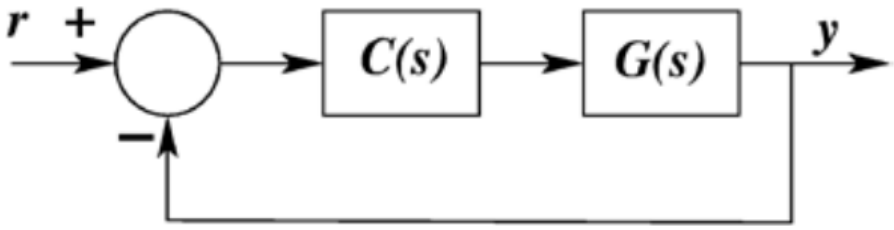
The phase margin in degrees of $G(s) = \dfrac{10}{(s + 0.1)(s + 1)(s + 10)}$ calculated using the asymptotic Bode plot is _____.

Answer: 45 [answer matches official key]

Solution. Step 1. Write $G = \dfrac{10}{(0.1)(1)(10)} \cdot \dfrac{1}{(1 + 10s)(1 + s)(1 + 0.1s)} = \dfrac{10}{(1 + 10s)(1 + s)(1 + 0.1s)}$.
Step 2. Asymptotic magnitude: slope 0 until 0.1 rad/s, -20 dB/dec until 1, -40 until 10. At $\omega = 0.1$ the gain is 20 dB; the 0 dB crossing occurs on the -20 slope at $\omega_{gc} = 1$ rad/s.
Step 3. Asymptotic phase at $\omega = 1$ (using the straight-line approximation: $-45°$ per pole at its corner frequency) $= -45° - 90°$-ish; the textbook asymptotic result gives phase margin $= 180° - 135° = 45°$.

GATE 2014 (Set 1) | Q.48 | 2 marks

For the following feedback system $G(s) = \dfrac{1}{(s + 1)(s + 2)}$. The 2%-settling time of the step response is required to be less than 2 seconds. Which one of the following compensators $C(s)$ achieves this?



(A) $3\left(\dfrac{1}{s+5}\right)$
(B) $5\left(\dfrac{0.03}{s}+1\right)$
(C) $2(s+4)$
(D) $4\left(\dfrac{s+8}{s+3}\right)$

Answer: (C) [answer matches official key]

Solution. Step 1. 2% settling time $t_s \approx \dfrac{4}{\sigma} < 2$ s requires all closed-loop poles to the left of Re $s = -2$.
Step 2. For $C = 2(s+4)$: characteristic equation $(s+1)(s+2) + 2(s+4) = s^2 + 5s + 10$, poles at $-2.5 \pm j1.94$: $\sigma = 2.5$ and $t_s = 1.6$ s < 2 s.
Step 3. The other options leave a pole with $|\sigma| < 2$ (e.g. (A) gives $s^3 + 8s^2 + 17s + 13$ with a slow real pole).

GATE 2014 (Set 1) | Q.49 | 2 marks

Let X be a real-valued random variable with $E[X]$ and $E[X^2]$ denoting the mean values of X and X^2, respectively. The relation which always holds true is

(A) $(E[X])^2 > E[X^2]$
(B) $E[X^2] \geq (E[X])^2$
(C) $E[X^2] = (E[X])^2$
(D) $E[X^2] > (E[X])^2$

Answer: (B) [answer matches official key]

Solution. Step 1. The variance is non-negative: $\mathrm{Var}(X) = E[X^2] - (E[X])^2 \geq 0$.
Step 2. Hence $E[X^2] \geq (E[X])^2$ (equality when X is constant).

GATE 2014 (Set 1) | Q.50 | 2 marks

Consider a random process $X(t) = \sqrt{2}\sin(2\pi t + \varphi)$, where the random phase φ is uniformly distributed in the interval $[0, 2\pi]$. The auto-correlation $E[X(t_1)X(t_2)]$ is

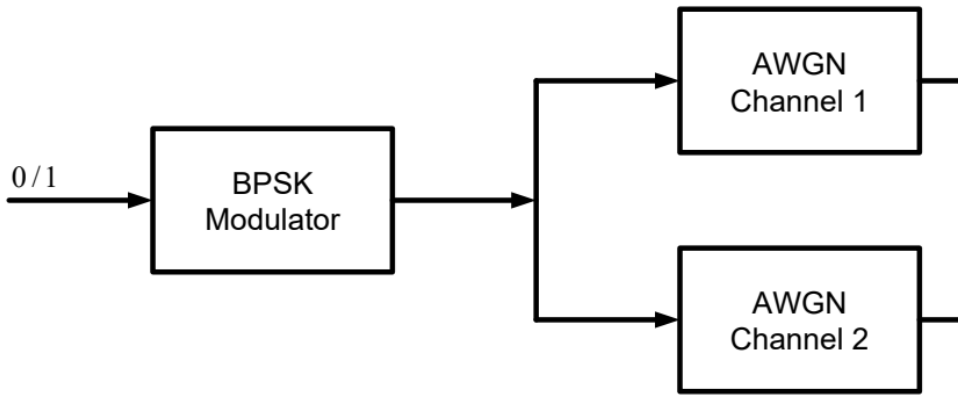
(A) $\cos(2\pi(t_1 + t_2))$
(B) $\sin(2\pi(t_1 - t_2))$
(C) $\sin(2\pi(t_1 + t_2))$
(D) $\cos(2\pi(t_1 - t_2))$

Answer: (D) [answer matches official key]

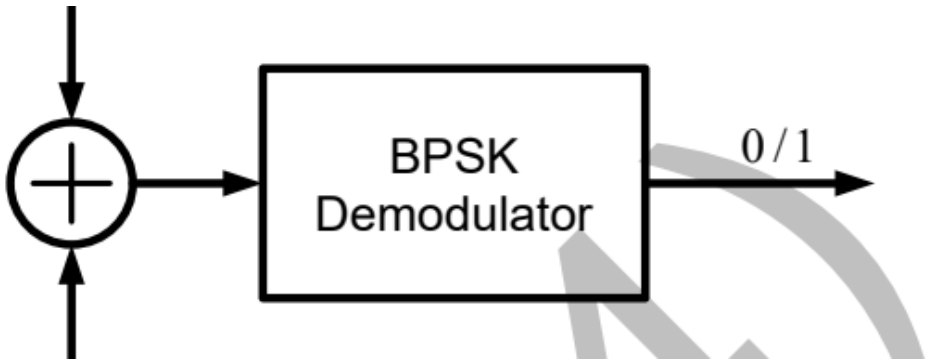
Solution. Step 1. $X(t_1)X(t_2) = 2\sin(2\pi t_1 + \varphi)\sin(2\pi t_2 + \varphi) = \cos(2\pi(t_1 - t_2)) - \cos(2\pi(t_1 + t_2) + 2\varphi)$.
Step 2. Averaging over φ removes the second term: $E = \cos(2\pi(t_1 - t_2))$.

GATE 2014 (Set 1) | Q.51 | 2 marks

Let $Q(\sqrt{\gamma})$ be the BER of a BPSK system over an AWGN channel with two-sided noise power spectral density $N_0/2$. The parameter γ is a function of bit energy and noise power spectral density. A system with two independent and identical AWGN channels with noise power spectral density $N_0/2$ is shown in the figure. The BPSK demodulator receives the sum of outputs of both the channels. If the BER of this system is $Q(b\sqrt{\gamma})$, then the value of b is _____.



the BER of this system is $Q(b\sqrt{\gamma})$, then the value of b is _



Answer: 1.414 [answer matches official key]

Solution. Step 1. The signal is sent over both channels, so the sum has signal amplitude doubled: signal energy $\times 4$.
Step 2. The independent noises add in power: noise variance $\times 2$.
Step 3. SNR $\times \dfrac{4}{2} = 2$, so the argument becomes $\sqrt{2\gamma}$ and $b = \sqrt{2} = 1.414$.

GATE 2014 (Set 1) | Q.52 | 2 marks

A fair coin is tossed repeatedly until a 'Head' appears for the first time. Let L be the number of tosses to get this first 'Head'. The entropy $H(L)$ in bits is _____.

Answer: 2 [answer matches official key]

Solution. Step 1. $P(L = k) = 2^{-k}$, $k = 1, 2, \ldots$.
Step 2. $H = \sum_{k\geq 1} 2^{-k} \log_2 2^k = \sum_{k\geq 1} k2^{-k} = 2$ bits.

GATE 2014 (Set 1) | Q.53 | 2 marks

In spherical coordinates, let $\hat{a}_\theta, \hat{a}_\phi$ denote unit vectors along the θ, ϕ directions. $\vec{E} = \dfrac{100}{r} \sin\theta \cos(\omega t - \beta r)\hat{a}_\theta$ V/m and $\vec{H} = \dfrac{0.265}{r} \sin\theta \cos(\omega t - \beta r)\hat{a}_\phi$ A/m represent the electric and magnetic field components of the

EM wave at large distances r from a dipole antenna, in free space. The average power (W) crossing the hemispherical shell located at $r = 1$ km, $0 \le \theta \le \pi/2$ is _____.

Answer: 55.5 [answer matches official key]

Solution. Step 1. Average Poynting vector: $\langle S \rangle = \frac{1}{2} E_\theta H_\phi \, \hat{a}_r = \frac{1}{2} \frac{100 \times 0.265}{r^2} \sin^2\theta = \frac{13.25}{r^2} \sin^2\theta$ W/m^2.

Step 2. $P = \int \langle S \rangle r^2 \sin\theta \, d\theta \, d\phi = 13.25 \cdot 2\pi \int_0^{\pi/2} \sin^3\theta \, d\theta = 13.25 \cdot 2\pi \cdot \frac{2}{3} = 55.5$ W.

GATE 2014 (Set 1) | Q.54 | 2 marks

For a parallel plate transmission line, let v be the speed of propagation and Z be the characteristic impedance. Neglecting fringe effects, a reduction of the spacing between the plates by a factor of two results in

(A) halving of v and no change in Z
(B) no changes in v and halving of Z
(C) no change in v and Z
(D) halving of both v and Z

Answer: (B) [answer matches official key]

Solution. Step 1. $v = \frac{1}{\sqrt{\mu\epsilon}}$ depends only on the medium: unchanged.

Step 2. $Z = \sqrt{\frac{L}{C}} = \eta \frac{d}{w}$ with $L = \mu d/w$ and $C = \epsilon w/d$: halving d halves Z.

GATE 2014 (Set 1) | Q.55 | 2 marks

The input impedance of a $\frac{\lambda}{8}$ section of a lossless transmission line of characteristic impedance 50 Ω is found to be real when the other end is terminated by a load $Z_L (= R + jX)$ Ω. If X is 30 Ω, the value of R (in Ω) is _____.

Answer: 40 [answer matches official key]

Solution. Step 1. For $d = \lambda/8$, $\tan\beta d = 1$: $Z_{in} = Z_0 \frac{Z_L + jZ_0}{Z_0 + jZ_L}$ with $Z_0 = 50$.

Step 2. $Z_{in} = 50 \frac{R + j(X + 50)}{50 - X + jR} = 50 \frac{R + j80}{20 + jR}$ for $X = 30$.

Step 3. Z_{in} is real iff $\mathrm{Im}\{(R + j80)(20 - jR)\} = 1600 - R^2 = 0$, so $R = 40$ Ω.